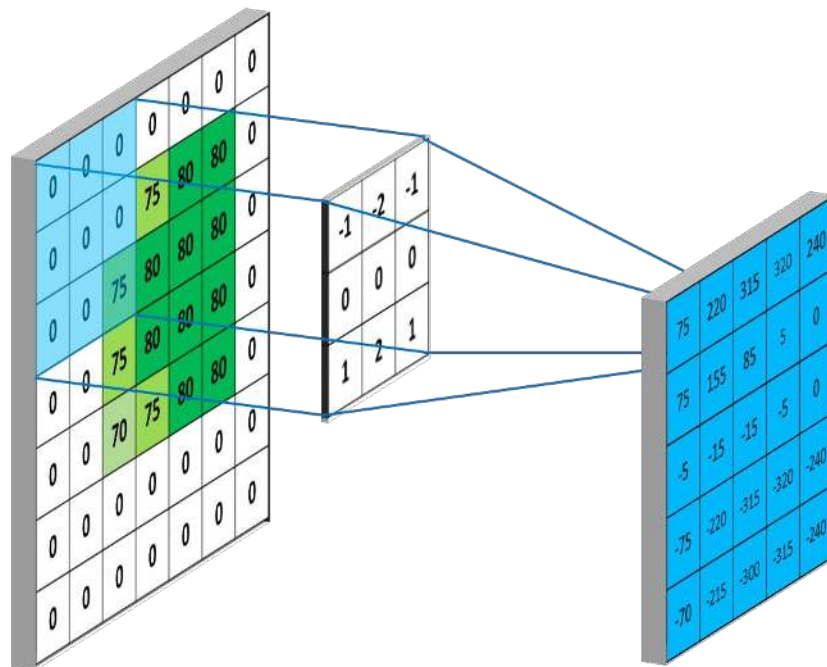


Improving classification performance of Convolutional Neural Networks by changing the convolutional kernel shape

Introduction

- In Convolutional Neural Networks, kernel are usually arbitrarily have the shape $N \times M$.
- Can we learn different kernel shapes from data to improve classification?



What has been done :

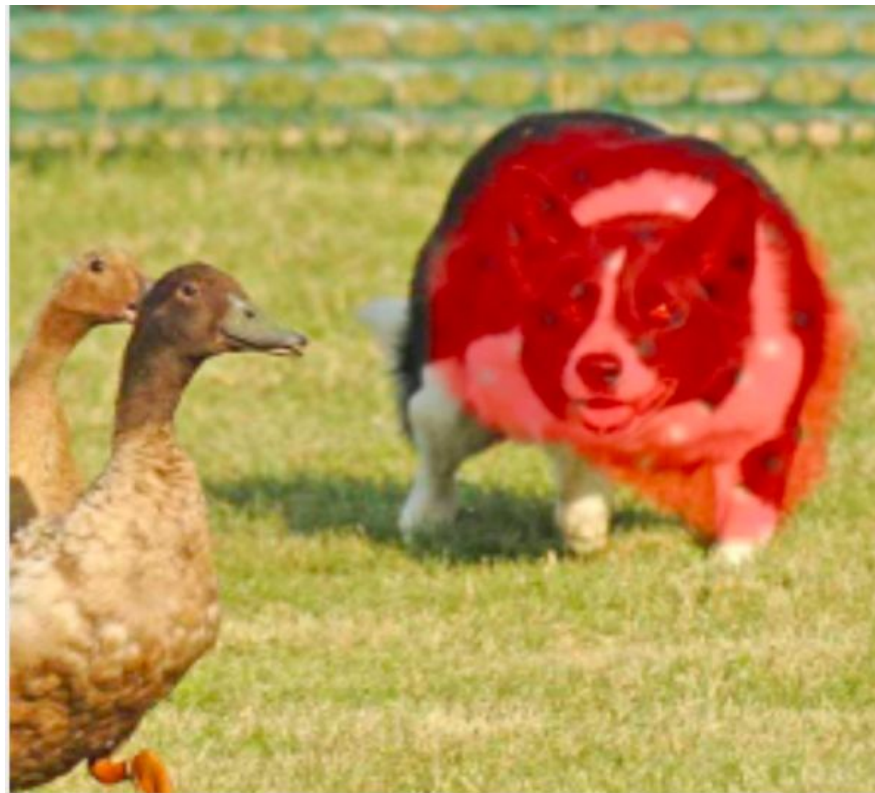
- Attempted to predict basic shapes in images using Residual Neural Networks.
 - But it used more memory than what was available.
- Implemented a Convolutional Neural Network instead.

What has been done :

- Introduced our own kernels as a first attempt.

Currently doing :

- Saliency map on images to see if it improves accuracy.



What's next :

- Learn different kernel shapes from the data at hand.
- See how it changes prediction accuracy.

Questions ?